Project 6: Multiply-Accumulate Module

Introduction

Multiply–accumulate (MAC) units form the computational core of modern AI hardware accelerators. They execute the repeated multiply and add operations that dominate neural network workloads. The efficiency of MAC units directly determines the overall speed and energy performance of deep learning systems. Designing optimized MAC hardware enables higher throughput and lower power consumption, making it essential for real-time and edge AI applications.

Project Description

In this project, you will design and implement a hardware multiply–accumulate (MAC) unit using SystemVerilog. You will explore various data representations, such as fixed-point, floating-point, or logarithmic formats, and evaluate their impact on accuracy, resource usage, and power. You will implement the datapath, control logic, and pipelining stages to improve performance and scalability. After simulation and verification, you will synthesize the design on an FPGA and measure its throughput, latency, and area.

Expected Results

You will produce a functional and verified MAC hardware module optimized for AI workloads. The design should demonstrate significantly higher performance and energy efficiency than software implementations, showcasing how custom hardware accelerators enhance the speed and scalability of deep learning computations.

Useful References

1. Multiply-Accumulate Operation Wikipedia
2. Low-Complexity Multiply-Accumulate Unit for Weight-Sharing Convolutional Neural Networks
3. Jack Unit: An Area- and Energy-Efficient Multiply-Accumulate (MAC) Unit Supporting Diverse Data Formats